The Exam is in **Three** Pages

Q(1) MULTIPLE CHOICE: From what you understand in class, choose the most fit answer and fill in the following table (Only one choice)

1	2	3	4	5

1. What is Artificial Intelligence?
   a. Putting your intelligence into Computer
   b. Programming with your own intelligence
   c. Making a Machine intelligent
   d. Putting more memory into Computer
2. If the state at the next instant in time depends only on the state at the current time, the task environment is said to be
   a. episodic
   b. stochastic
   c. dynamic
   d. deterministic
3. What is state space?
   a. The definition of a problem
   b. Problem you design
   c. Representing your problem with variable and parameter
   d. A space where you know the solution.
4. Which search method takes less memory?
   a. Depth-First Search
   b. Breadth-First search
   c. A*
   d. Hill Climbing
5. The minimax algorithm in Game playing is:
   a. Best move algorithm
   b. Local search algorithm
   c. Greedy method
   d. Depth first algorithm

CamScanner

Question 5: [5 points] Chose *the one* answer that best answers the question

1) In which of the following search algorithms is it possible to confuse a local maximum with a global maximum?
   a. depth first
   b. best first search
   c. hill climbing
   d. greedy search

2) An optimal search algorithm
   a. finds the solution that has the lowest path cost among all solutions
   b. finds all solutions
   c. finds a solution using the least amount of memory
   d. is guaranteed to find a solution when there is one

3) What is the main difference between the minimax and alpha-beta methods?
   a. only minimax is guaranteed to find the best move for Max
   b. alpha-beta pruning only works for randomly ordered successor nodes
   c. alpha-beta pruning typically examines fewer nodes than minimax
   d. minimax only works if the whole search tree can be computed before the respective best moves for Max and Min are determined

4) What is the key distinguishing characteristic of a learning agent?
   a. A learning agent can make a distinction between world states.
   b. A learning agent maps a real number (level-of happiness) to each state
   c. A learning agent knows how to relate an action to a goal.
   d. A learning agent tries to improve its performance by suggesting improvements after observing its own actions and their outcomes.

5) What constitutes a full solution to a constraint satisfaction problem?
   a. the set of all variables used in the specification of the problem
   b. the set of all variables together with their respective potential values
   c. an assignment of values to some (but not all) variables
   d. an assignment of values to all variables

CamScanner

Question (2) (15 points)

A. (1 point) What is the definition of an admissible heuristic?

B. (4 points) Multiple Choice:

1. Which of the following algorithms is optimal
   a. Bidirectional
   b. Iterative deepening Algorithm
   c. Hill Climbing
   d. Both (a) and (b)
   e. None of the above

2. $\exists x(S(x) \wedge C(x))$ is equivalent to which of the following?
   a. $\exists x(C(x) \wedge S(x))$
   b. $\exists x S(x) \wedge \exists x C(x)$
   c. $\sim\forall x(S(x) \rightarrow \sim C(x))$
   d. Both (a) and (c)
   e. All the above

Scanned with CamSca

3. How could you translate "There are at most two apples" into FOL
   a. $\forall x \forall y \forall z\ ((Apple(x) \wedge Apple(y) \wedge Apple(z)) \rightarrow (x=y \vee y=z \vee x=z))$
   b. $\forall x \forall y\ (Apple(x) \wedge Apple(y)) \rightarrow (x=y \vee y=x)$
   c. $\exists x \exists y\ (Apple(x) \wedge Apple(y) \wedge \sim(x = y) \wedge \forall z\ (Apple(z) \rightarrow (x = z \vee y = z))$
   d. Both (a) and (c)
   e. None of the above

4. How could you translate "There are exactly two apples" into FOL
   a. $\exists x \exists y\ ((Apple(x) \wedge Apple(y) \wedge \forall z\ Apple(z)) \rightarrow (x=z \vee y=z))$
   b. $\exists x \exists y\ (Apple(x) \wedge Apple(y) \wedge \sim(x = y) \wedge \forall z(Apple(z) \rightarrow (x=z \vee y=z)))$
   c. $\exists x \exists y\ (\sim(x = y) \wedge \forall z(Apple(z) \leftrightarrow (x=z \vee y=z)))$
   d. $\exists x \exists y\ (Apple(x) \wedge Apple(y) \wedge \sim(x = y)) \wedge \forall x \forall y \forall z((Apple(x) \wedge Apple(y) \wedge Apple(z)) \rightarrow (x=y \vee y=z \vee x=z))$
   e. Both (b) and (c) and (d)
   f. All the above

CamScanner

Q(1) True or false questions:

a. (T / F) The iterative deepening search algorithm is optimal.

b. (T/ F) The number of correctly placed tiles in the 8-puzzel is an admissible function.

c. (T/ F) h=0 in the 8-puzzel is an admissible function.

d. (T/ F) Local beam search retains the k best successors of the k states in the previous step

CamScanner

B. (8 points) Just write True/False:

i. (T) The omniscience of an intelligent agent requires the agent to have infinite knowledge.

ii. (F) Depth-first search always expands at least as many nodes as A* search with an admissible heuristic.

iii. (F) True $\models$ False.

iv. (F) Percept based intelligent agent collects information from its Actuators.

v. (T) $(A \leftrightarrow B) \wedge (\neg A \vee B)$ is satisfiable

vi. (T) Forward Chaining with the Generalized Modus Ponens is complete in First order predicate Calculus.

vii. (T) A Unifier is a substitution for variables in atomic sentences that makes them identical

viii. (F) An existentially quantified sentence with one variable can be instantiated several times as long as the Skolem constants used are all distinct.

CamScanner

Question (3) (15 points)

A. (6 points) Just write True/False:

i. (T…..) $\exists \mathbf{x}\ \forall y$ equal(x,y) is unsatisfiable

ii. (F…..) Depth-first search always expands at least as many nodes as Depth limited search.

iii. (T…..) $(A \leftrightarrow B) \wedge (\neg A \vee B)$ is satisfiable

iv. (T…..) Forward Chaining with the Generalized Modus Ponens is complete in First order predicate Calculus.

v. (T…..)Prolog works with backward chaining.

vi. (T…..)A Unifier is a substitution for variables in atomic sentences that makes them identical

CamScanner

B. **(8 points)** Circle T or F for True/False: **(two circles is a wrong answer)**

i. (T / F) Iterative Deeping Search Algorithm is optimal.

ii. (T / F) An agent is usually operating in a partially observable environment.

iii. (T / F) A* Search cannot be used in robotics because percepts, states, and actions are continuous.

iv. (T / F) Depth-first Iterative deepening always returns the same solution as breadth-first search if b is finite.

v. (T / F) The clause (~A∨~B∨C) entails the clause (~A ∨~B)

vi. (T / F) A Sound inference rule can derive all sound sentences from a knowledge base

vii. (T / F) Context-free grammars may fail to capture English grammar because the meaning of an English sentence may depend on the context of the utterance.

viii. (T / F) Ambiguity in natural language can always be resolved by checking dictionary definitions of words

CamScanner

- **Question 2:**
- **(T/ F) why?**If *h* were not admissible in A*, the optimality of the search may change.
- Yes! It can make the good optimal goal look as though it is very far o_, and take you to a suboptimal Goal
- **(T/ F) why?**The rational agent is guaranteed to reach an optimal solution.
- T: Not omnisci.